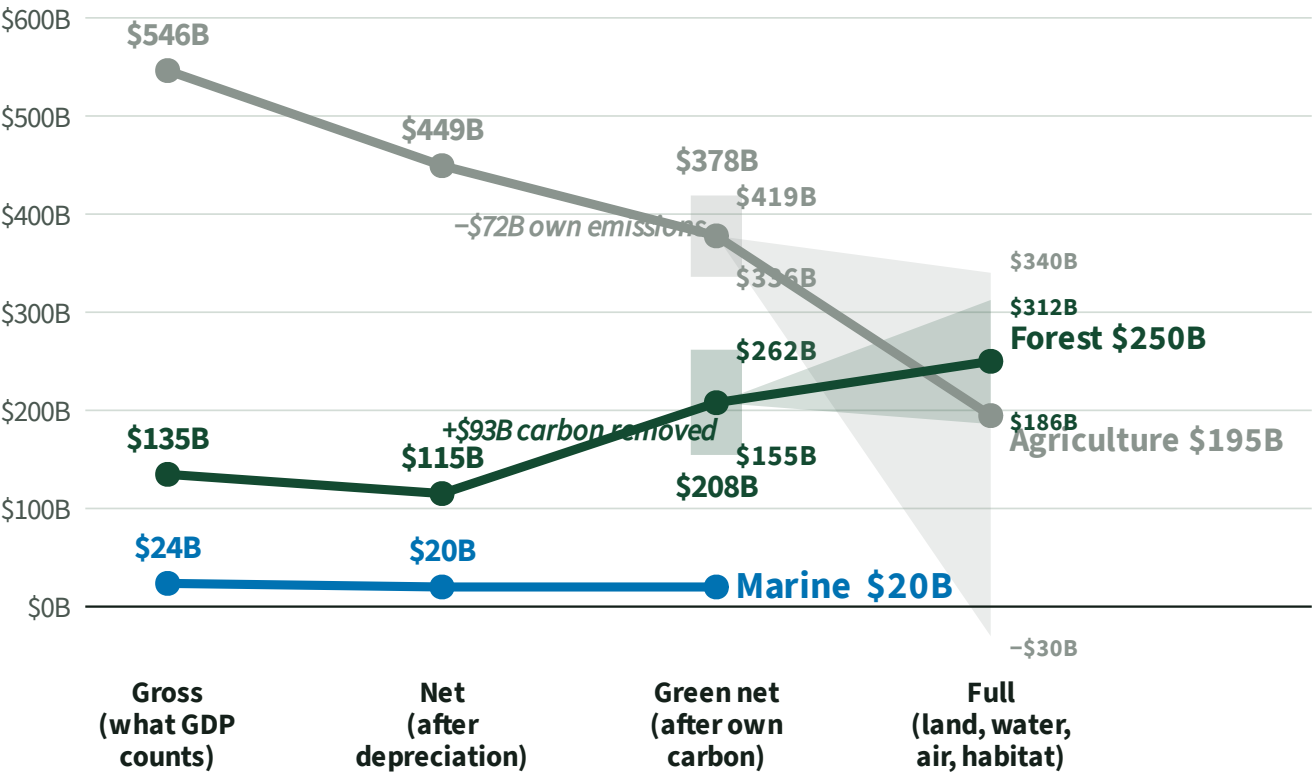


America's forests are undervalued, misunderstood and underinvested

With carbon counted, forests are about half the size of agriculture, not a quarter. Charge farms for land, nitrogen and water damages, credit forests for clean air, and at the central carbon price forests come out ahead (central 128%, lower bound 55%), so they could rival agriculture, with mill costs and forest water and habitat still uncounted. The country grows far more wood than it cuts yet imports more than it sells, and funds forestry research at a seventh of agriculture's level.

1. Undervalued

Value added to US GDP, 2022, \$B, by accounting basis

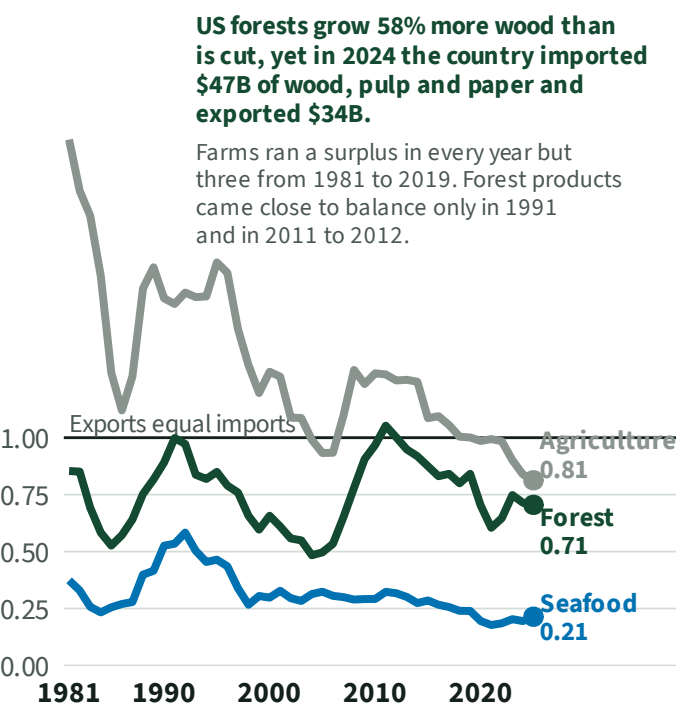


Full step: agriculture charged \$183B for land, nitrogen and water damages (FAO 2023); forest credited \$42B for clean air (Cavender-Bares et al. 2022), with forest water and habitat still unpriced and mill emissions and effluent not charged, so the comparison is indicative, not a bound.

Panel 1. Sectors are primary production plus first stage manufacturing: farms plus food, beverage and tobacco manufacturing (BEA 110C, 3110, 3120); wood and paper manufacturing (3210, 3220); marine living resources (BEA MESA). Forestry and logging and farm support services are excluded from both land sectors (BEA reports them in one line, 113F). Net value added subtracts BEA current-cost depreciation by detailed industry (September 30, 2026; BEA rates the detail below its aggregates); marine depreciation is a proxy. Shaded fans are scenario ranges, not statistical confidence intervals: each sector's own carbon (forest removal 771.7 MMT CO₂e, EPA GHGI 2024; direct farm emissions 596 MMT, EPA 2022) at \$51, \$120 and \$190 per t (IWG 2021, EPA 2023). BEA publishes no sampling error for value added. Fourth step, agriculture: FAO State of Food and Agriculture 2023, US environmental hidden costs of agrifood systems in 2020 (million 2020 PPP dollars): land 114,931, nitrogen 62,215, blue water 6,018 (climate 53,142 omitted because the carbon step already charges farm emissions), uncertainty minus 57% to plus 100%. Fourth step, forest: air quality regulation by US trees, 37% of \$114B (range \$85B to \$137B, 2010 USD), Cavender-Bares et al. 2022, PLOS Sustainability and Transformation; carbon and wood products from that study are already counted. Forest water yield (drinking water for about 180 million Americans) and habitat have no verified national dollar value, and mill emissions and effluent are not charged, so the forest full value is indicative, not a bound. Food plant and fishing fleet emissions are not netted. Panel 2. UN Comtrade, SITC Rev. 1 for 1981 to 1990 and HS for 1991 to 2025 (1991 overlap within 0.05); exports free on board, imports free on board where reported, otherwise CIF; agriculture HS 01 to 24 less seafood, forest products HS 44, 47, 48, seafood HS 03, 1604, 1605. FAOSTAT forestry trade shows primary wood and paper products in deficit every year from 1961 to 1990. Growth to harvest ratio from Arkansas Center for Forest Business (2023). Panel 3. Hatch and McIntire-Stennis FY2018 (OMB, COSSA); Sea Grant base \$65.0M plus aquaculture \$11.5M, FY2018 enacted (Consolidated Appropriations Act 2018 explanatory statement), whisker to FY2024 (\$91.5M). The target is half of Hatch, the green net (carbon counted) forest to agriculture ratio in panel 1. Chart: Weiskittel and Daigneault, University of Maine.

2. Misunderstood

Exports per dollar of imports, 1981 to 2025



3. Underinvested

Federal research formula funds, FY2018

